Quant Rule — Architecture Characteristics

Rating: 1 = very low requirement, 5 = very high requirement.

1. Overall cost

Characteristic	Level	Kata	Why it matters	Target	Evidence
Overall cost	2/5	R1-R6	the distributed split must stay cheap to build and run	at most 9 containers, 0 managed services, 1 host	deployment inventory

2. Structural architecture characteristics

Characteristic	Level	Kata	Why it matters	Target	Evidence
Simplicity	5/5	R1-R6	hidden complexity hides errors	at most 5 configuration files; no infrastructure	file count
Modularity	4/5	R1, R2, R4	data, signals, hypothesis, evaluation change independently	0 cyclic dependencies	dependency check
Maintainability	4/5	R1-R4	one correction must not touch unrelated parts	at most 1 module and 2 files per fix	change size
Testability	5/5	R4, R5, R6	results must be checked against known answers	at least 90% coverage of statistics; suite at most 5 minutes	coverage and suite time
Configurability	4/5	R1, R2, R3	range, signals, costs, thresholds are set per hypothesis	0 code changes to change a hypothesis	configuration diff

3. Engineering architecture characteristics

Characteristic	Level	Kata	Why it matters	Target	Evidence
Evolvability	4/5	R1, R2	new hypotheses and signals arrive continuously	at most 1 day per new signal	change size
Deployability	2/5	R6	a result must reproduce elsewhere	1 command, 0 manual steps, at most 15 minutes	timed install

4. Operational architecture characteristics

Characteristic	Level	Kata	Why it matters	Target	Evidence
Performance	3/5	R2, R4	large history, batch experiments	at most 60 s, 4 GB RAM, 80% of one core	benchmark
Reliability	4/5	R4, R5, R6	completed results must not be corrupted	50 of 50 experiments at 80% CPU for 1 hour	constrained run
Responsiveness	2/5	R2	exploration is interactive	at most 2 s per query	timed query
Scalability	2/5	R2, R4	one user, growing experiment count	10 to 100 experiments, every result correct	run count
Elasticity	1/5	-	no burst expected	1 user, at most 2 s, at most 70% resources	peak test
Fault tolerance	2/5	R6	an interrupted experiment restarts	100% of completed results valid after a kill	kill test